

VIS 2019

VitalVizor: A Visual Analytics System for Studying Urban Vitality

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Urban Vitality

- Urban design is about place-making [Buchanan 1988, p.33].



Urban Design Process

- Vitality is the first basic elements of good urban form [Lynch's 1981, p. 113].



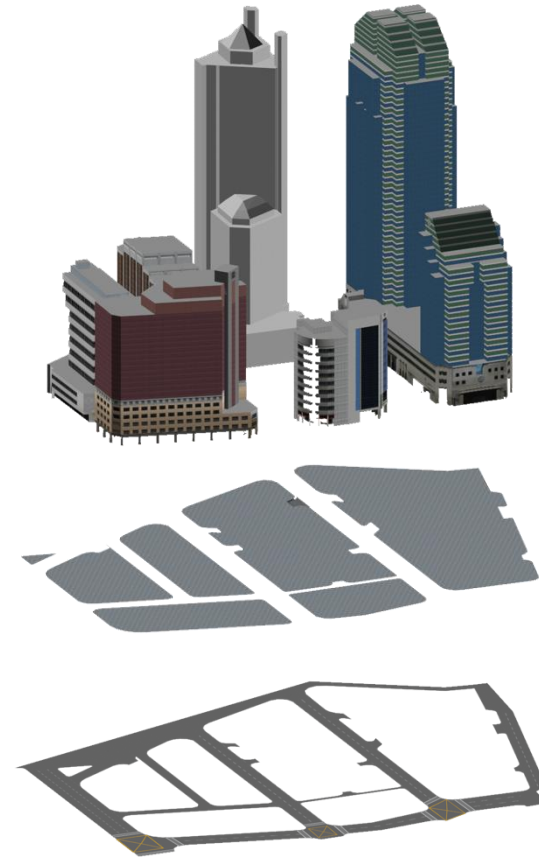
Urban design is essential about
improving urban vitality!

Urban Vitality

- The term 'urban vitality' is implicit and vague
 - *Diversity* [Jacobs, 1961]
 - 1) *Visual*, 2) *functional*, 3) *environmental*, and 4) *vibrant urban experiences*, or *urbanity* [Cook, 1980]
 - Urban composition not only *functions* properly but *pleasing in appearance* [Rowley, 1994]
- A gap between theory and practice
 - From qualitative descriptions to quantitative metrics
 - A visual analytics system
 - Locate which areas are vital/unvital
 - Identify what factors make the places vital/unvital

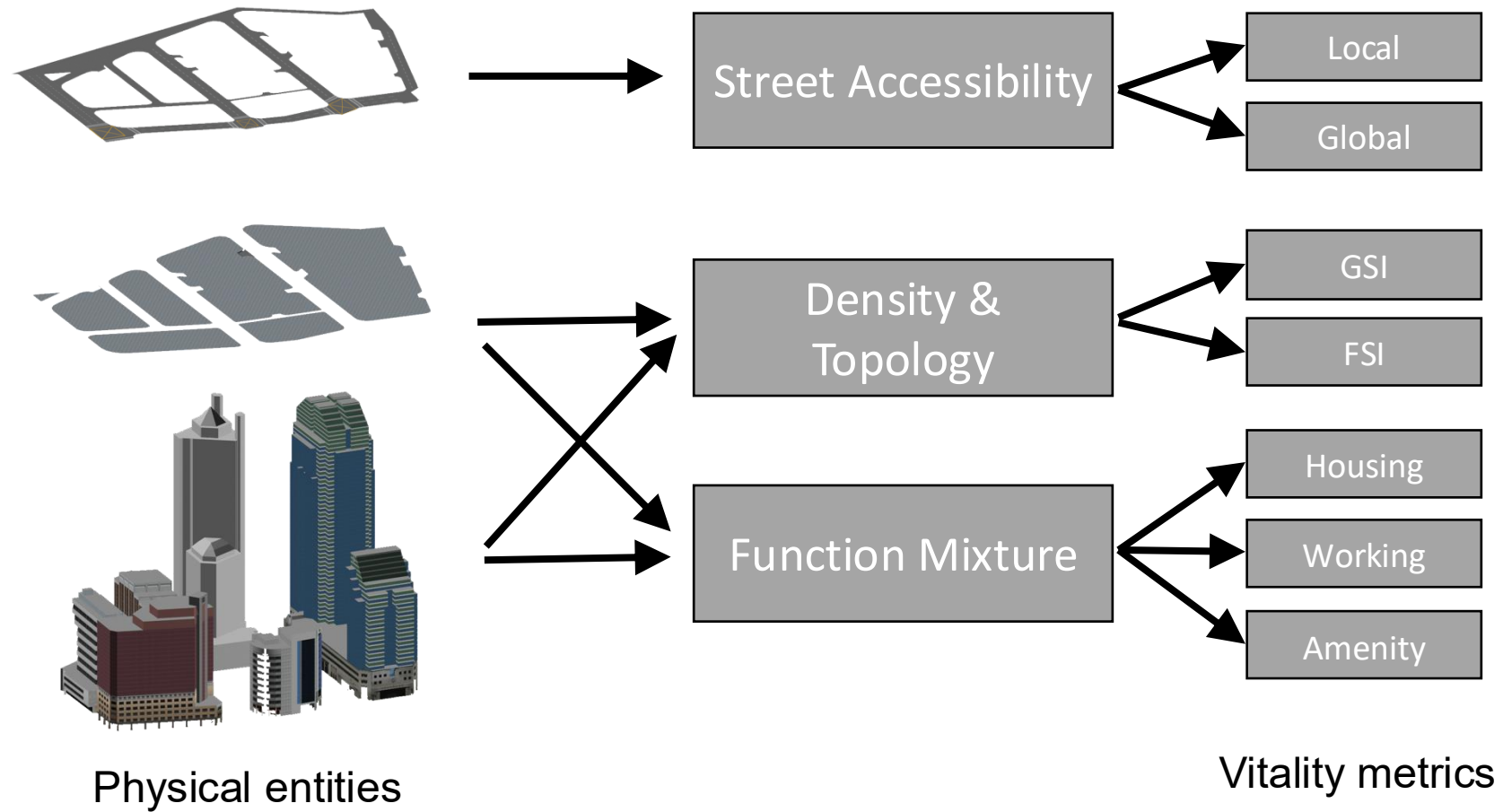
Vitality Metrics

- Quantitative metrics for 'urban vitality'



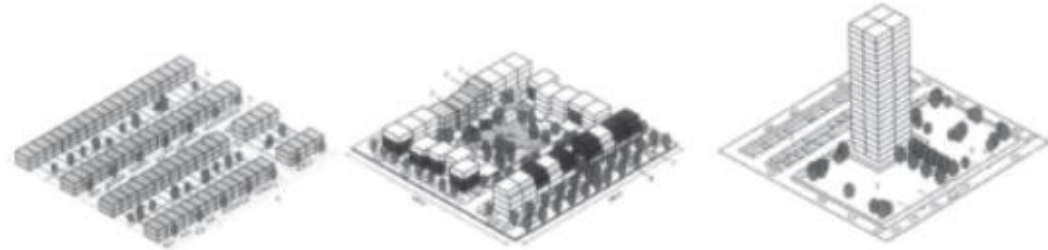
Vitality Metrics

- Quantitative metrics for 'urban vitality'



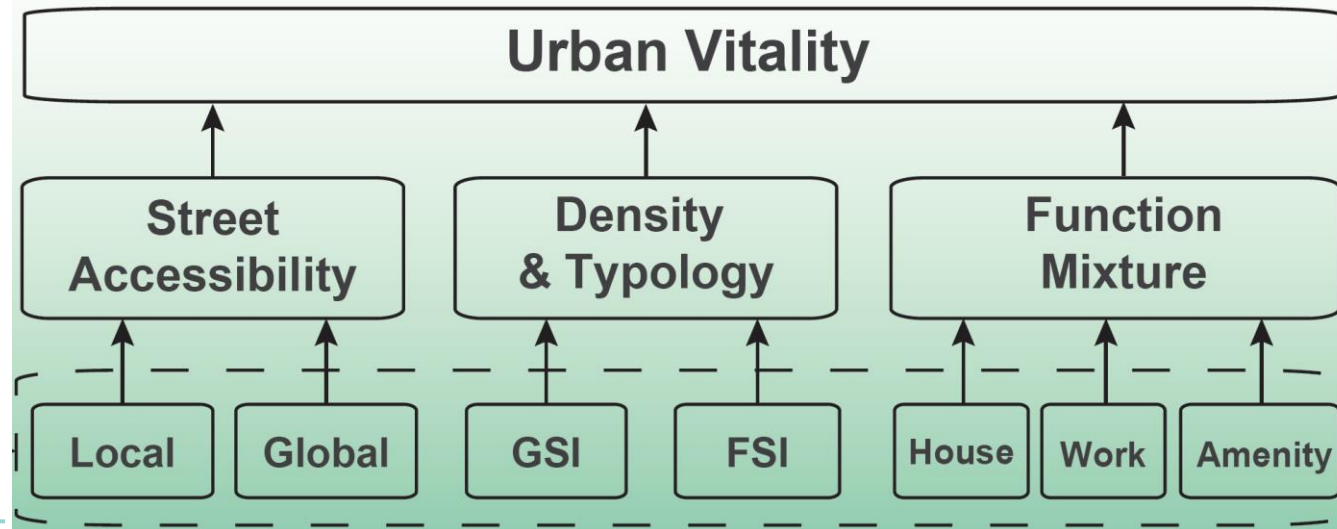
Vitality Metrics

- Street accessibility (<https://sdna.cardiff.ac.uk/sdna/>)
 - Mean Angular Distance (MAD) $SAD(x) = \sum_{y \in R_x} d\theta(x, y) P(y)$
 - Betweenness centrality
- Density & topology
 - Ground space index (GSI): GSI
 - Floor space index (FSI): FSI =
- Function mixture
 - Housing, working, and amenity
 - Mono-functional, bi-functional, mixed, and highly-mixed



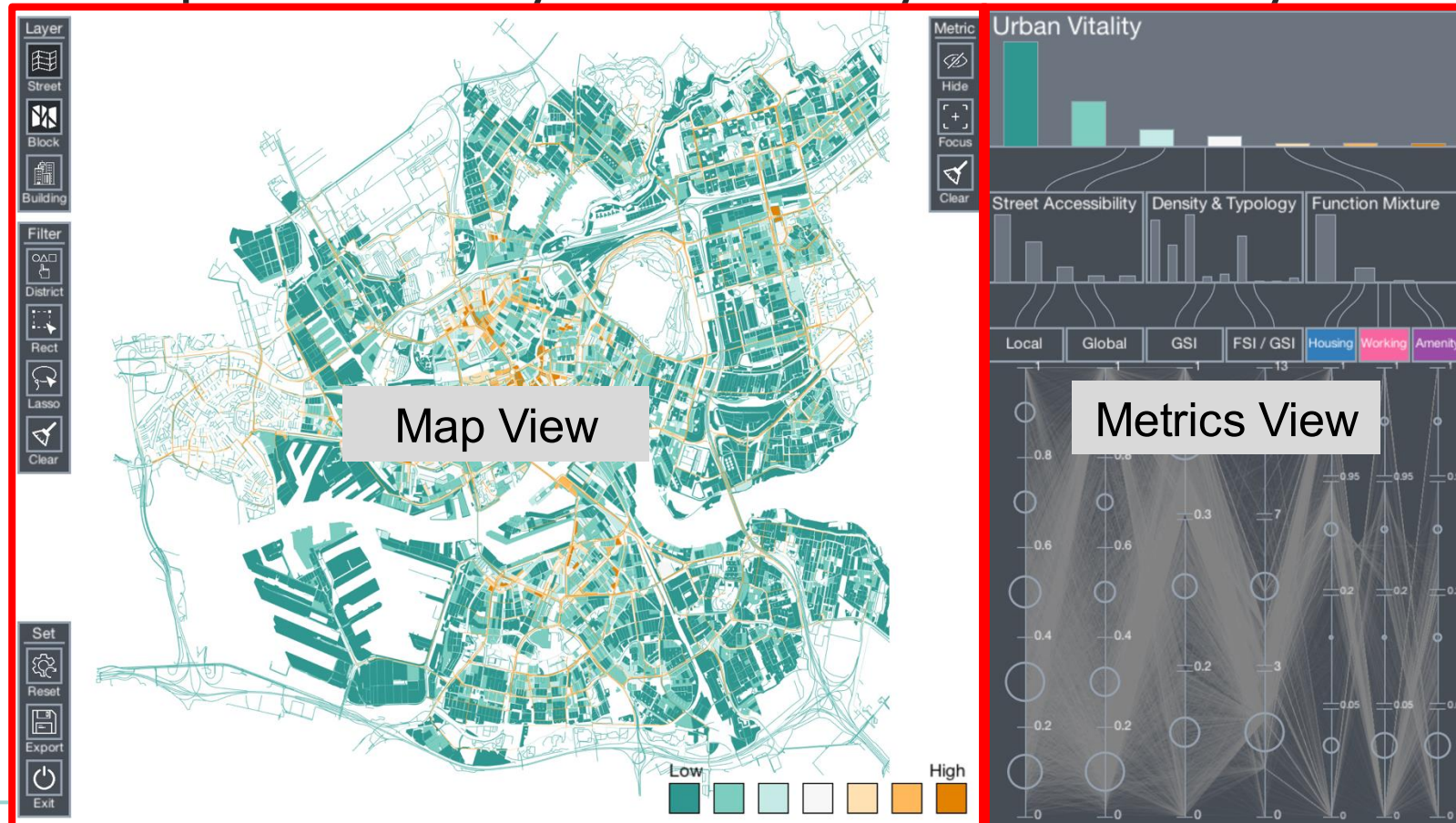
Data Characterization

- Physical entities by nature exhibit ***geospatial*** information
 - Buildings → 3D
 - Blocks → 2D, multiscale
 - Street networks → 2D, graph
- Vitality metrics are ***multi-dimensional*** and ***hierarchical***



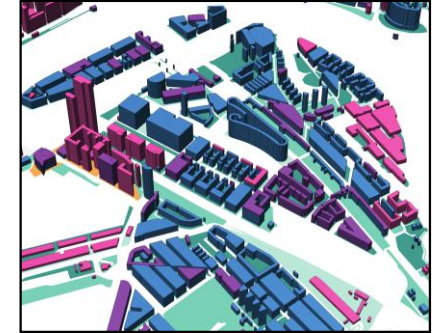
VitalVizor System

- We develop VitalVizor system to study urban vitality.



Map View

- Design choices
 - Street network → 2D polyline
 - Blocks → 2D polygon
 - Buildings → 3D
- Color scheme: streets and blocks
- Interaction
 - Map navigation
 - Selection by district, lasso, or rectangle

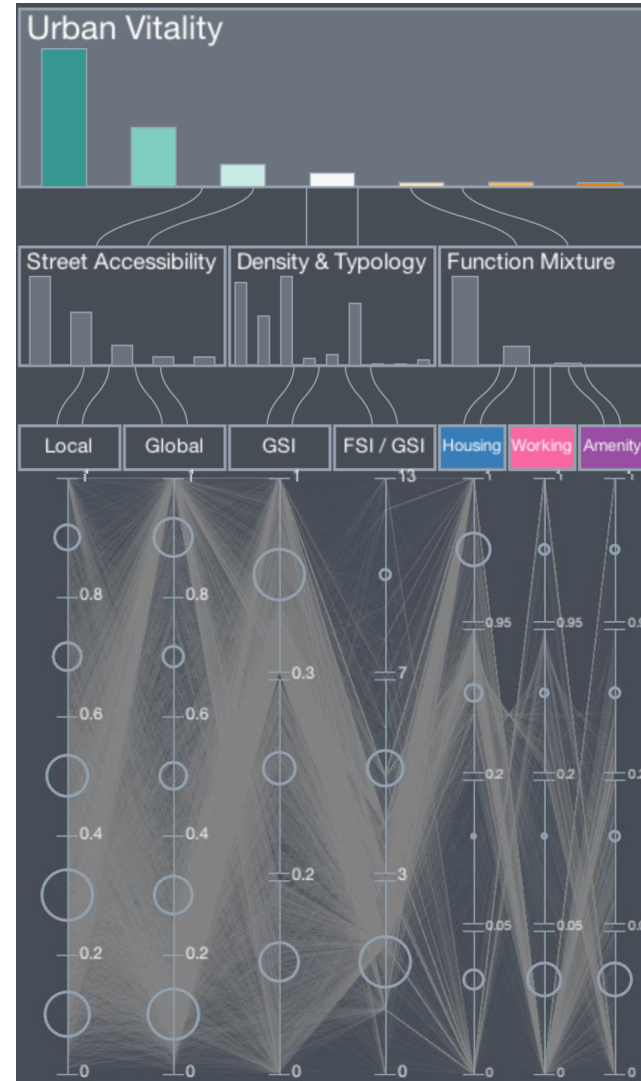


■ Housing ■ Working ■ Amenity



Metrics View

- Design choices
 - Hierarchical relationship
→ tree-structure diagram
 - High-dimensional information
→ PCP
- Interaction:
 - Filtering



Layer

Street

Block

Building

Filter

District

Rect

Lasso

Clear

Set

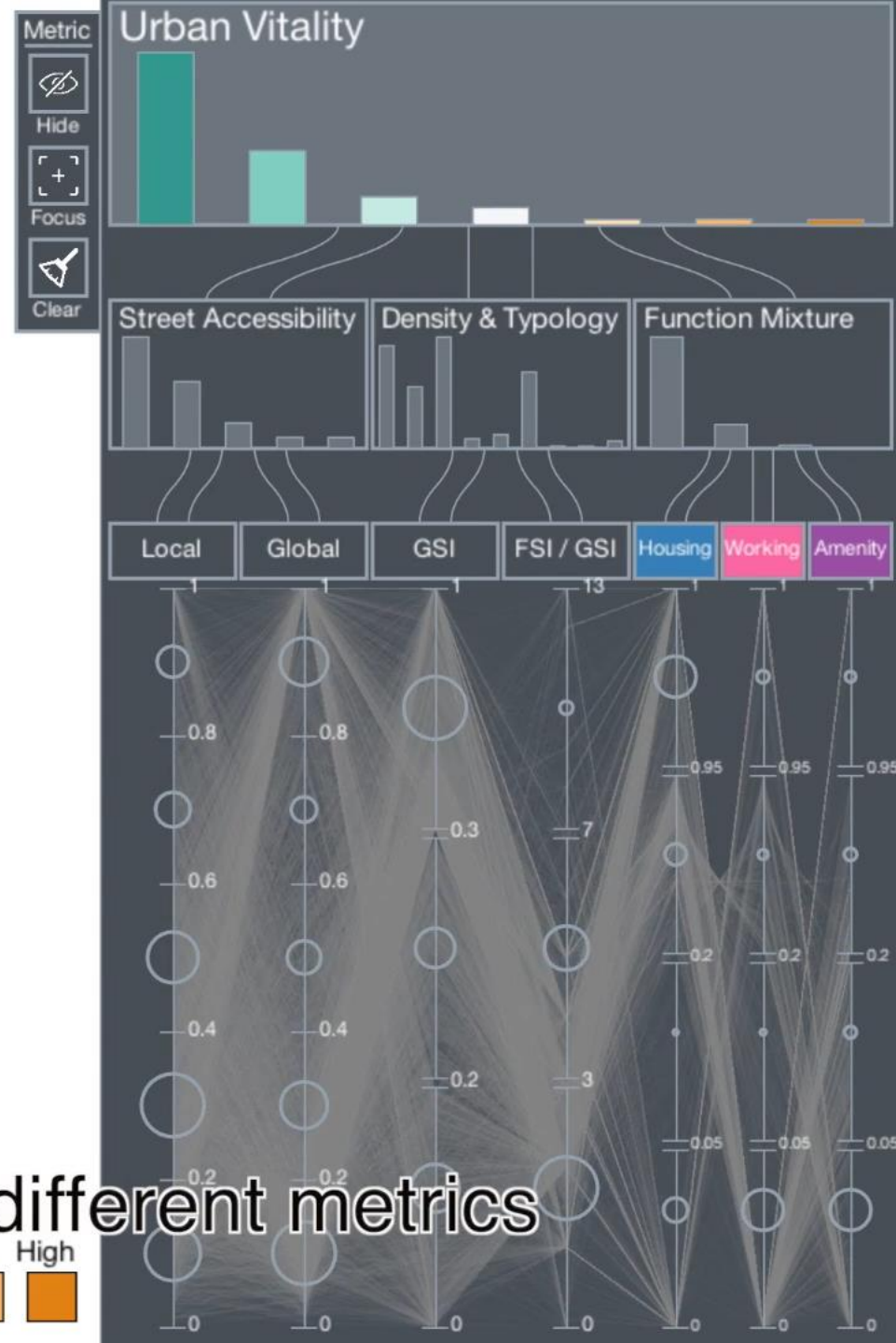
Reset

Export

Exit



In Metric View, users can select different metrics

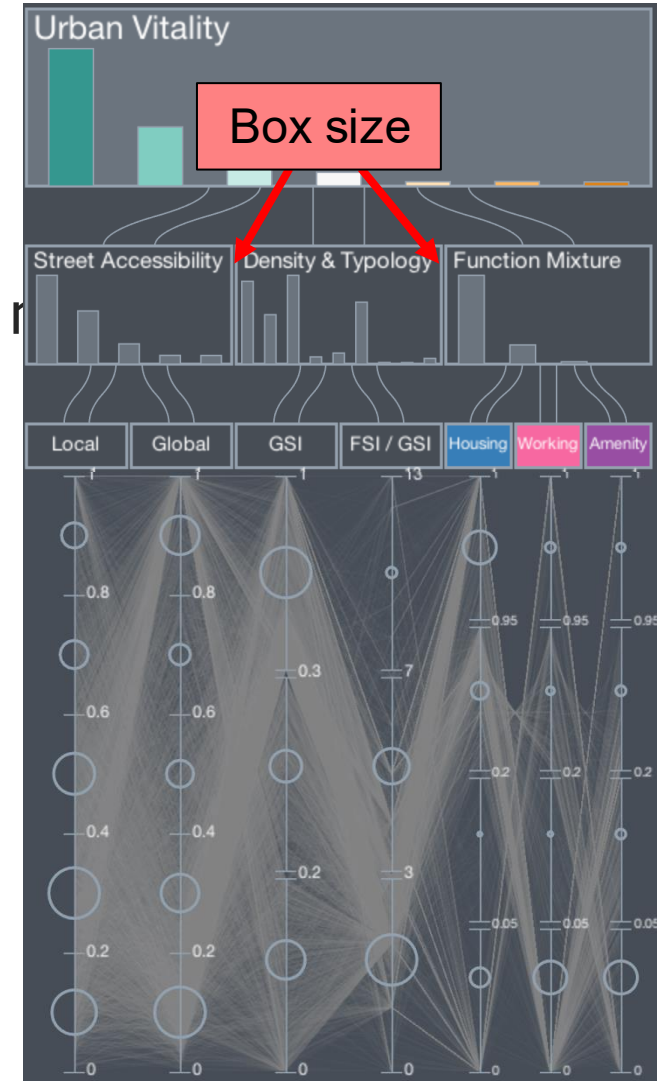




Demonstration speaks
louder than conversation!

Takeaways

- Not much to take away
 - Linked views for exploring physical entities and vitality
 - Combination of tree diagram and PCP
- More functionalities like dragging.
- But I do realize:
 - Visualization design and city design are similar
 - Visualization field is at the early stage



谢谢!

Thank
You!

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